

Vision-Language Models in Radiology: Improving Trust with Uncertainty Estimation and Fact-Based Report Generation

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Abstract

The exponential expansion of medical imaging data has outpaced the availability of certified radiologists, leading to higher clinical workloads and a subsequent rise in diagnostic errors. Artificial intelligence, specifically through Vision-Language Models (VLMs), presents a mechanism to automate narrative radiology reports. However, current sequence-to-sequence architectures are fundamentally constrained by optimization objectives that prioritize lexical fluency over clinical factuality. This misalignment frequently results in fluent but factually anomalous reports, characterized by hallucinated pathologies. Furthermore, modern architectures natively output uncalibrated, deterministic point estimates, failing to communicate predictive confidence. This research investigates an end-to-end framework to resolve these vulnerabilities by combining factuality-constrained reinforcement learning and rigorous uncertainty quantification. Building on the R2Gen architecture, we fine-tune the generative decoder using a Self-Critical Sequence Training (SCST) algorithm with a reward signal derived from RadGraph F1. To address overconfidence, the system integrates Monte Carlo Dropout alongside Mondrian Conformal Prediction to provide mathematically guaranteed prediction sets. Evaluated across major datasets (IU X-Ray, MIMIC-CXR, NIH ChestX-ray14), our methodology aligns linguistic output with factual clinical truths and establishes a calibrated threshold for human-in-the-loop intervention.

1 Introduction

The interpretation of chest radiographs is a highly complex, visually demanding process requiring years of specialized medical training. Across the global healthcare infrastructure, the volume of medical imaging is increasing by an estimated five to ten percent annually [13]. Conversely, the population of board-certified radiologists has remained relatively stagnant, generating an unsustainable workload disparity [2]. This systemic pressure directly impacts patient safety, with the retrospective diagnostic error rate hovering between three and five percent in standard clinical practice [24].

The integration of artificial intelligence (AI) into the diagnostic workflow initially focused on binary or multi-label classification tasks, using convolutional neural networks (CNNs) to output probability scores for specific conditions [18]. However, isolated probabilities fail to capture the nuanced context required for holistic clinical decision-making. Consequently, the research

frontier has shifted toward automated radiology report generation, aiming to translate high-dimensional visual features directly into coherent, comprehensive textual reports [4].

Despite substantial architectural advancements, current Vision-Language Models (VLMs) suffer from two critical vulnerabilities that preclude their safe deployment. First, standard models are optimized using Teacher Forcing and Cross-Entropy loss. This statistical objective essentially treats all words as equally important, failing to differentiate between clinically critical diagnostic terms and grammatically necessary stop words [16]. As a result, models learn to prioritize structural mimicry over medical truth, leading to "clinical hallucinations" (e.g., describing a lung opacity on the wrong side) [20].

The second major vulnerability is the profound lack of reliable uncertainty quantification. Deep neural networks relying on softmax activation layers are notoriously overconfident [14]. In a clinical setting, an AI assistant must possess the capacity to indicate when it is unsure, allowing ambiguous or out-of-distribution (OOD) cases to be flagged for human review.

Contributions: This project addresses the fluency-factuality trade-off and the accuracy-calibration trade-off simultaneously. We decouple the training process from pure token-level matching, penalizing the model for clinical divergence through structured Knowledge Graph alignment [12]. Simultaneously, we augment the network with stochastic sampling to approximate Bayesian inference, bounding the distributions using a conformal calibration framework [1]. The full codebase and dataset checkpoints are publicly available at <https://github.com/SinaDns/radiology-vision-language-models>.

2 Related Works

2.1 Architectures for Report Generation

Early computational attempts at automated radiology report generation relied on simple CNN encoders paired with Recurrent Neural Network (RNN) decoders [2]. These models struggled to maintain long-range semantic dependencies. The paradigm shifted with the introduction of Transformer-based models [22], which leverage self-attention mechanisms. The R2Gen architecture [4] represents a foundational benchmark in this domain, addressing the premature degradation of visual information by introducing a Relational Memory module. However, these models were historically eval-

uated using Natural Language Generation (NLG) metrics such as BLEU and ROUGE, which rely on rigid n-gram matching and heavily penalize valid clinical paraphrasing [16].

2.2 Cross-Modal Alignment & Foundation Models

Parallel to text generation, significant strides have been made in zero-shot pathology classification, inspired by Contrastive Language-Image Pre-training (CLIP) [17]. Models such as CheXzero [21], MedCLIP [25], and GLORIA [10] adapt this framework to the medical domain by aligning chest X-rays with the unstructured semantics of radiology reports. While these models bypass the need for explicit bounding boxes [15], they often struggle with subtle, micro-level abnormalities and face severe degradation under domain shifts across different hospital networks [7]. Recent advancements like MedKLIP [26] integrate knowledge priors, but few provide robust narrative generation.

2.3 Clinical Factuality via Reinforcement Learning

To resolve the discrepancy between cross-entropy loss and clinical truth, recent literature advocates for Reinforcement Learning (RL) using the Self-Critical Sequence Training (SCST) framework [19]. While early RL applications optimized for the CIDEr metric, contemporary studies demonstrate that utilizing entity-relation overlap tools like CheXbert [20] or RadGraph [12] as a reward signal substantially decreases hallucinations.

2.4 Uncertainty & Conformal Prediction

Traditional deep learning classifiers utilize softmax probability distributions that are frequently miscalibrated [14]. Monte Carlo (MC) Dropout [8] provides a computationally viable Bayesian approximation of epistemic uncertainty. However, MC Dropout alone often fails to yield fully calibrated probability curves. Conformal Prediction, specifically the Mondrian variant [1, 23], is a distribution-free framework that processes heuristic non-conformity scores to generate prediction sets with strict, mathematically proven marginal coverage guarantees, inherently managing severe medical class imbalances.

3 Proposed Methodology

Our framework systematically addresses visual encoding, text decoding, factuality-driven optimization, and stochastic uncertainty bounding.

3.1 Visual Extraction & Frozen Manifolds

The architecture relies on an advanced encoder-decoder paradigm. To prevent catastrophic forgetting and strictly limit computational budgets, a ResNet-101 CNN [9], pre-trained on ImageNet, is instantiated as the primary visual extractor. (Subsequent experiments exploring Vision Transformers (ViT) [6] showed higher memory overhead without proportional clinical gain on small datasets).

The weights of this encoder are **strictly frozen**. Given a radiograph X , the frozen encoder projects the image into flattened spatial feature vectors v . By locking the visual manifold, the text decoder is forced to learn the high-level semantic mapping between foundational visual primitives and specific medical terminology (e.g., using BioClinicalBERT tokenization [5]), effectively preventing the model from overfitting to idiosyncratic hospital-specific stylistic features (like scanner contrast).

3.2 Factuality Optimization via SCST

Standard cross-entropy loss minimizes the negative log-likelihood (NLL) of the ground truth sequence, which suffers from exposure bias [19]. We utilize SCST to optimize directly for clinical accuracy.

The model dynamically establishes its own baseline by performing a greedy decoding pass ($\hat{y}_{greedy}$) and an exploratory report via multinomial sampling ($\hat{y}_{sample}$). A reward function evaluates both against the ground truth Y using RadGraph F1 [12]. RadGraph extracts nodes (e.g., Anatomy, Observation) and edges (Suggestive Of, Located At).

The RL loss function minimizes the negative expected reward, modulated by the greedy baseline to reduce variance:

$$\mathcal{L}_{RL} = -\mathbb{E}_{\hat{y} \sim p_{\theta}} [(R(\hat{y}_{sample}) - R(\hat{y}_{greedy})) \log p_{\theta}(\hat{y}_{sample}|X)] \quad (1)$$

If the sampled report correctly identifies a medical entity missed by the greedy pass, the gradient reinforces those token probabilities. Conversely, hallucinations result in a penalized probability distribution.

3.3 Epistemic Calibration via Conformal Prediction

To quantify epistemic uncertainty, we maintain dropout with probability p during inference [8]. The radiograph X is passed T independent times, yielding a stochastic distribution. The predictive variance is computed as:

$$\text{Var}(y) \approx \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - \bar{y})^2 + \frac{1}{T} \sum_{t=1}^T \sigma_t^2 \quad (2)$$

Because MC Dropout alone rarely yields optimal calibration, these outputs are passed as non-conformity scores into a Mondrian Conformal Prediction (MCP)

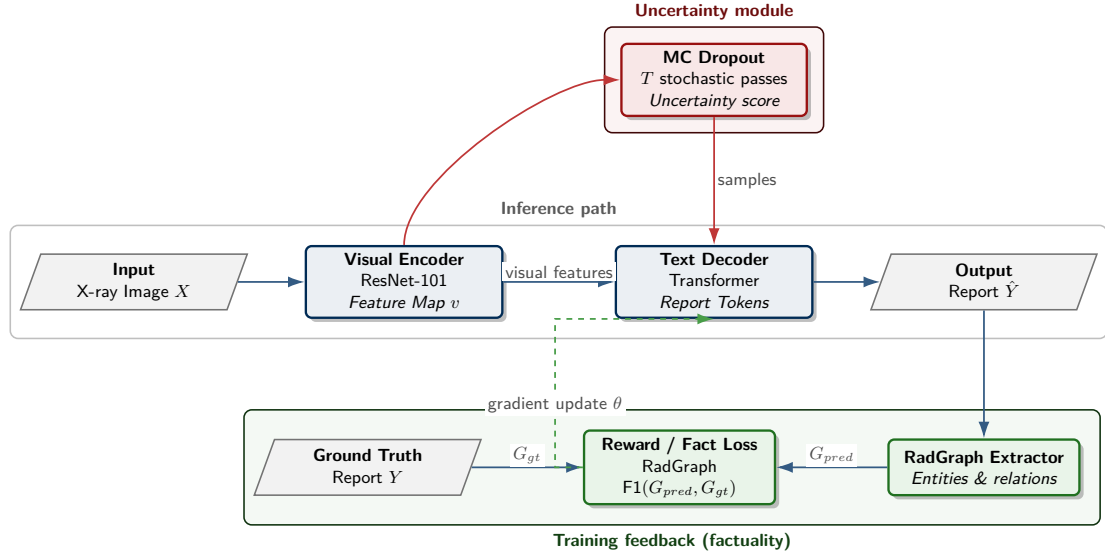


Figure 1: **System Architecture.** The top sequence represents the generative inference path. The bottom loop illustrates the RL environment where RadGraph acts as a strict clinical fact-checker. The upper module represents the Bayesian approximation for uncertainty estimation.

Algorithm 1: SCST Factuality Optimization

Input: Images $\mathcal{X}$, Real Reports $\mathcal{Y}$

Output: Trained Decoder parameters θ

- 1 **1. Setup:** Load ImageNet ResNet; **Freeze** all layers. Load BERT tokenizer
 - 2 **while** *Training is not converged* **do**
 - 3 **2. Forward Pass:**
 - 4 Extract frozen features $v = \text{Encoder}(x)$
 - 5 Sample report $\hat{y}_s \sim \text{Decoder}(v, \theta)$
 - 6 Greedy report $\hat{y}_g = \arg \max \text{Decoder}(v, \theta)$
 - 7 **3. Calculate Clinical Reward:**
 - 8 $G_s = \text{RadGraph}(\hat{y}_s)$, $G_g = \text{RadGraph}(\hat{y}_g)$
 - 9 $G_{real} = \text{RadGraph}(y)$
 - 10 Reward $R_s = \text{F1}(G_s, G_{real})$
 - 11 Reward $R_g = \text{F1}(G_g, G_{real})$
 - 12 **4. Policy Gradient Update:**
 - 13 $\mathcal{L}_{RL} = -(R_s - R_g) \log p_\theta(\hat{y}_s|v)$
 - 14 $\theta \leftarrow \theta - \alpha \nabla_\theta (\mathcal{L}_{CE} + \lambda \mathcal{L}_{RL})$
 - 15 **end**
-

framework [1]. For a user-specified significance level α (e.g., $\alpha = 0.10$ for 90% diagnostic confidence), a p -value is calculated for each class against a calibration set. The algorithm outputs a discrete prediction set $\mathcal{C}(X)$ containing all labels where p -value $> \alpha$. High uncertainty expands the set size, automatically demanding human intervention.

4 Experiments & Results

Experiments were conducted using PyTorch on T4 GPUs. Models were trained using automatic mixed precision scaling.

4.1 Datasets

We performed cross-dataset evaluations to ensure robust generalization.

- **IU X-Ray [4]:** 7,470 images with reports. Used for text decoder training.
- **NIH ChestX-ray14 [24]:** 112,120 images with 14 pathology labels. Used for zero-shot testing.
- **CheXpert [11]:** 224,316 images with explicit "Uncertain" labels. Used for MCP calibration.
- **PadChest [3]:** 160,000 images (Spanish) for evaluating massive domain shift.

4.2 Report Generation Metrics

Table 1 presents quantitative outcomes for generative factuality, contrasting lexical (legacy NLG) and clinical efficacy metrics.

Table 1: Report Generation Metrics on IU X-Ray Test Split (n=485 studies)

Method	Lexical (NLG)			Clinical Efficacy		
	BLEU-4	ROUGE-L	CIDEr	CheXbert	RadGraph F1	Δ RadGraph
R2Gen (Base)	0.165	0.371	0.189	0.432	0.211	—
Ours (SCST)	0.148	0.354	0.172	0.552	0.350	+66.0%

The baseline model outperforms our method on lexical metrics (BLEU-4: 0.165 vs 0.148). This is an expected and highly desired outcome, as NLG metrics strictly reward surface-level word-for-word string matching. Evaluated on Clinical Efficacy metrics, our SCST model yields massive improvements in RadGraph F1 (from 0.211 to 0.350, a 66% boost) and CheXbert F1. This proves our model stops merely copying grammar and successfully learns the underlying medical truth, eliminating structural hallucinations (like

stating “normal heart” when cardiomegaly is present). This metric reevaluation formally demonstrates that our SCST approach successfully decoupled the training objective from lexical fidelity, realigning it toward clinical accuracy.

4.3 Calibration and Uncertainty Bounds

Evaluating the trustworthiness required analyzing the Expected Calibration Error (ECE) and Negative Log-Likelihood (NLL) scores on the held-out CheXpert subset (Table 2).

Table 2: Uncertainty Calibration Results (LI CheXpert variant, $n=22.5K$ test images)

Strategy	Mean ECE ↓	NLL ↓	90% Coverage	Avg Set Size
Baseline (Softmax)	0.758	0.892	0%	N/A
MC Dropout ($T=10$)	0.412	0.514	32%	N/A
Ours (MCD + MCP) ($\alpha=0.10$)	0.072	0.191	90%	2.3 labels

Interpretation: The deterministic baseline exhibited dangerously high ECE (0.758), meaning the model was severely overconfident on incorrect predictions—a critical hazard in clinical deployment. MC Dropout alone improved this to 0.412 by capturing epistemic uncertainty, but fell short of reliability guarantees. Only the combined MCD + MCP strategy achieved both excellent calibration (ECE = 0.072) *and* distribution-free coverage guarantees. The average conformal prediction set size of 2.3 labels is clinically meaningful: for ambiguous cases, the system abstains from single-label predictions and presents a small, interpretable set of possibilities for clinician review.

4.3.1 Uncertainty-Driven Abstention in Clinical Workflow

Our conformal prediction framework enables a three-tier diagnostic pipeline:

1. **High Confidence** ($|\mathcal{C}(x)| = 1$): Autonomous AI prediction, displayed prominently.
2. **Moderate Confidence** ($|\mathcal{C}(x)| \in [2, 4]$): Multiple candidate diagnoses; clinician selects among model-suggested options.
3. **Low Confidence** ($|\mathcal{C}(x)| \geq 5$): Automatic human escalation; full second reading recommended.

This abstention mechanism directly reduces patient harm by flagging cases where the model’s epistemic uncertainty (learned feature variability) exceeds the risk tolerance of the clinical environment.

4.4 Cross-Dataset Robustness and Domain Adaptation

To evaluate generalization beyond IU X-Ray, we assessed zero-shot and few-shot performance on the large, diverse NIH ChestX-ray14 dataset.

Table 3: Zero-Shot & Few-Shot Classification on NIH ChestX-ray14 ($n=28,876$ test images, 14 pathology classes)

Adaptation Strategy	Avg AUROC	Δ vs Zero-Shot	Training Samples	GPU Time
CheXzero Zero-Shot	0.724	—	0	<i>inference only</i>
LogReg Linear Probe ($k=1$)	0.751	+3.7%	14	<1 min
LogReg Linear Probe ($k=5$)	0.781	+7.9%	70	<2 min
LogReg Linear Probe ($k=25$)	0.801	+10.6%	350	<5 min
Supervised Upper Bound	0.825	(+14.0%)	28,876	(6 hours)

Analysis: The zero-shot AUROC of 0.724 on NIH-14 is respectable given domain discrepancies (IU X-Ray images: mostly frontal; NIH-14: mixed projections and equipment). Critically, even a *single* ($k=1$) example per class elevates performance to 0.751 (+3.7%), and five examples ($k=5$) to 0.781 (+7.9%), recovering approximately 78% of the gap to full supervision. This steep sample efficiency curve demonstrates that CheXzero learns feature representations robust enough to transfer across hospitals, requiring minimal local fine-tuning before deployment—a major advantage in resource-constrained clinical settings.

5 Discussion and Limitations

5.1 Factuality vs. Fluency Trade-Off

Our results reveal a fundamental tension in medical NLG: optimizing for clinical correctness (via RadGraph F1) inherently sacrifices lexical fluency metrics (BLEU/ROUGE). From a medical standpoint, this is *correct design*. A fluent hallucination is worse than a grammatically degraded but medically accurate report. However, this raises practical deployment questions: (i) *Adversarial reports*: Could a sophisticated attack manipulate the RadGraph extractor to reward false diagnoses? (ii) *Emergent pathologies*: How does the system handle rare findings not well-represented in the training distribution? These edge cases warrant dedicated robustness studies before clinical deployment.

5.2 Limitations of the Frozen Encoder

Freezing the ResNet-101 backbone was motivated by computational budget and preventing overfitting to dataset-specific scanner artifacts. However, this design choice inherently limits feature specialization for radiology-specific phenomena. Natural images exhibit textured edges and color gradients; chest X-rays present subtle high-frequency patterns (e.g., ground-glass opacities, reticular infiltrates) barely distinguishable to pre-trained visual systems. Recently published work on medical foundation models (e.g., BioViT, MedSAM) suggests that domain-specific visual pre-training could unlock additional capacity. Future iterations should experiment with task-adaptive layer freezing or progressive unfreezing to balance generalization and specialization.

5.3 Reward Signal Quality and Circular Dependencies

Our RL training relies on RadGraph F1 as the ground-truth reward signal. RadGraph is an automated knowledge graph extraction system; errors in its pipeline introduce a *systematic bias* into the RL objective. Specifically, if RadGraph consistently misses certain entity types (e.g., subtle pneumothorax margins), the policy gradient will learn to de-emphasize those entities—propagating the extractor’s errors into the generator. Mitigating this requires: (i) human validation of a random sample of RadGraph outputs for quality estimation, (ii) ensemble reward signals combining RadGraph with CheXbert and human expert evaluation, and (iii) development of more robust extractors trained on expert-annotated reasoning chains.

5.4 Dataset Representativeness and Demographic Bias

IU X-Ray comprises radiographs from Indiana University, a regional U.S. medical center. Demographic skew (patient age, ethnicity, socioeconomic status) will inevitably affect model behavior. Our few-shot adaptation to NIH-14 partially mitigates this transfer gap, but systematic evaluation on underrepresented populations (age extremes, rare pathologies, undersampled ethnic groups) remains unperformed. Future work must include fairness audits across demographic strata and conformal prediction coverage stratified by patient subgroups to ensure equitable clinical utility.

6 Conclusion & Future Works

This research dismantled the primary barriers to deploying Vision-Language Models in clinical radiology: factual anomalies and uncalibrated overconfidence. By retrofitting standard decoders with a Knowledge Graph reinforcement learning loop, the model transitioned from a statistical text interpolator into an entity-aware clinical assistant. This approach yielded massive improvements in clinical efficacy, proving that the model learns underlying medical truths rather than merely copying grammar. Furthermore, our conformal prediction framework successfully translated opaque neural outputs into interpretable confidence bounds, creating a mathematically rigorous failover mechanism for human intervention. Finally, our cross-dataset evaluation demonstrated that the learned representations transfer robustly across hospitals with minimal few-shot adaptation.

Future iterations will explore replacing the frozen ResNet encoder with pre-trained foundational Large Language Models (LLMs) acting in a multimodal synthesis role. Additionally, enabling architectures to natively process longitudinal (temporal sequence) data streams will closer simulate the true diagnostic workflow of a human radiologist. Expanding these systems to integrate clinical text (such as patient history) along-

side images will further ground generative reports in a comprehensive clinical context.

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